



PRE-GEN

Technical Specification

Valerii Egorov (Pastheroz)

Independent Researcher

ORCID: 0009-0000-2599-0547

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Abstract

Takedown requests, copyright suits, and content provenance standards act after content is generated. Emerging standards such as RSL and the Human Consent Standard let rights-holders publish machine-readable permissions and clearance paths for AI use. None of them gives a provider a signed answer for one specific generation request, bound to that request and to the evidence of what followed. Meanwhile, generative systems can depict or embody any entity to which rights attach: persons, estates, brands, synthetic characters, voices, virtual personas, and entity classes not yet anticipated.

I present PRE-GEN (PG), an open pre-generation authorization protocol for generative systems. PG introduces three mechanisms that close this gap: (1) a two-party cryptographically signed generative-rights license; (2) a pre-generation authorization query that AI providers execute before producing any output, answered with a signed decision — allow, not blocked, review, or deny; and (3) a pre-emptive opt-out that, once in effect, takes priority over every license in the registry that holds it. A subject's refusal terminates generation before any license is consulted, and no license can override it. After an allowed generation the provider files a receipt, and uses that need no license are reported as observations. Every decision is written to a signed, hash-chained audit log whose roots are periodically timestamped outside the registry. Version 5 fixes the identifier format, separates usage records from licenses, and lets several independent registries issue codes that never collide. Reconciling rights across registries is left to later versions. PG checks declared authority against one registry's records and conditions; it does not decide whether a use is lawful. The specification, byte-level test vectors, and conformance runner are published under Apache-2.0.

Keywords: pre-generation authorization, generative rights, two-party signed license, opt-out registry, signed decisions, PG codes, federated registries, pre-gen

1. Problem Statement and Scope

Generative AI systems can produce content depicting or embodying entities to which rights attach: natural persons, estates, brands and trademarks, synthetic characters, voices, and virtual personas. Takedowns, lawsuits, and provenance records all act after generation; published permission declarations (§1.1) say what a rights-holder allows in general, not whether one request is covered.

The cycle. A rights-holder registers a subject → licenses define what is permitted, to whom, for which use → before generating, a provider asks the registry and receives a signed decision → after generating, the provider reports what it produced → every step is written to a verifiable history. Sections 5–7 and 10 specify this cycle; §§8.2, 8.3 and 9 describe registry policy that varies between registries; §15 is future work and not part of the protocol. This paper explains the architecture and is not sufficient on its own to implement PG. Version 5 consists of this paper, the normative texts that fix every field and rule (PRE-GEN-SPEC.md, PG-CODE.md, REFUSAL-CODES.md), and the test vectors that fix byte-level results, published together [7]. The normative texts define the requirements in BCP 14 language (MUST, SHOULD, MAY), and the vectors test them. A disagreement between text and vector is an erratum to be corrected, never a rule, and this paper yields to both.

PRE-GEN (PG) is a pre-generation authorization protocol for generative systems. It introduces three key mechanisms:

Two-party signed generative-rights license. Every license is cryptographically signed by the rights-holder and countersigned by the registry operator. Neither signature alone makes a license. Where the rights-holder holds its own key (self custody), a compromised registry operator cannot issue a license without the rights-holder's signature. Where the registry holds the key on the rights-holder's behalf (managed custody), that guarantee rests on the registry's key vault and its record of the instruction (§3).

Pre-generation authorization query. AI providers query the registry before generating any content and receive a signed decision. The decision precedes the first pixel. Where provenance standards describe what was generated, PG answers, before generation, whether the registry's records permit it.

Pre-emptive opt-out with priority over licenses. A subject can refuse all generation before anyone attempts to obtain a license. The opt-out is checked before license resolution and cannot be overridden by any license in the registry that holds it; its exact reach is stated in §8.1.

Contribution

PG addresses a gap that existing standards were not designed for. Content provenance systems describe what was generated; PG answers whether a registry's records permit it before it is generated. Authentication protocols establish identity; PG records generative rights — what a specific party is authorized to generate depicting a specific subject, under what conditions, with what restrictions. The two-party signed license, the pre-generation query, the opt-out with priority over licenses, the four-layer architecture, and a registry model in which any number of operators can issue compatible identifiers together constitute pre-generation authorization for generative AI systems.

Scope. PG standardizes compliant generation flows. It does not and cannot prevent unauthorized generation by uncooperative providers, local open-source inference, or offshore services. The protocol operates within the compliant ecosystem: providers who choose to integrate. A signed decision records what the registry answered to a declared request; it does not prove that the output matched the request, that every generation was reported, or that a use is lawful.

1.1 Related standards

RSL [8] lets publishers declare machine-readable licensing terms for AI crawling, training, and use, discovered through robots.txt, HTTP headers, or feeds, with license tokens obtained from a license server and reused across requests. **The Human Consent Standard** [9], an RSL extension published as a draft in August 2026, declares AI restrictions and clearance paths for identities, characters,

works, and marks, optionally certified by a third party. A declaration can send an AI system to an external server for automated authorization. Both describe what a rights-holder permits and where to obtain permission; neither specifies a signed answer to one specific generation request. PG is complementary: it answers one declared generation request with a signed decision bound to that request, its user, and its intended use, and links that decision to what the provider reported afterwards. A future bridge could let an HCS clearance path lead to a PG registry. It would require the registry to implement the clearance-server interface that HCS defines, which this version neither specifies nor implements.

	RSL 1.0 [8]	HCS 1.0 draft [9]	PRE-GEN v5
How permission is obtained	OAuth token from a license server (§5.4)	a declaration may name a license server (§3.1.10); its protocol is not yet specified	one query per generation request (§7)
What the answer is bound to	a resource; reused until it expires	not specified	subject, prompt hash, model, intended use, user
Per-request answer verifiable offline	no: opaque token, checked by the server (§5.5)	no per-request answer specified; declarations can be certified with JWS (§3.3)	yes: Ed25519 signature, published keys
After use	reports required when the license includes <reporting>; format left to profiles (§3.12)	not specified	receipt per allow, observations, signed audit log
Refusal over permission	prohibitions precede permissions (§3.6)	conflicts read as prohibited (§3.1.9.7)	opt-out over every license, within one registry (§8.1)

2. Protocol Layers

The protocol is organized in four layers. Layer 1 is the foundation; Layers 2–4 are optional extensions built on it — the provenance bridge, for example, points to Layer 1's decision and receipt. A provider that implements only Layer 1 is already useful, and the extensions add capability without modifying it.

Layer 1 — Core Authorization Protocol. Two-party signed licenses, a verification API returning signed decisions, receipts filed after generation, and an append-only, hash-chained audit log with external anchoring. This is the minimum viable protocol.

Layer 2 — Identity Resolution. Determines whether a prompt references a registered subject. Each registry publishes a public subject index; detection runs on the provider's side and produces *candidates*, never proof. Providers may use name matching, embedding classifiers, voice-print or face matchers, or third-party entity linking without modifying the authorization layer. Authorization is always asked for a named subject.

Layer 3 — Policy Profiles. Jurisdiction- and business-dependent rules: registration trust, review of commercial requests, territorial restrictions, legal overrides, and dispute flows are expressed as registry policy, not as core protocol logic.

Layer 4 — Provenance Bridge. The registry issues a signed assertion (`pg.assertion.v1`) that a provider embeds in its own C2PA manifest, pointing to the decision and receipt behind a file. Layer

4 is useful but not required for the authorization guarantee; an end-to-end integration with a C2PA manifest has not yet been tested.

3. Trust Model

Every subject is bound to an Ed25519 signing key. The key may be held by the subject (self custody) or held encrypted by the registry and used only on the subject's instruction (managed custody). The two modes give different guarantees. Under self custody the subject signature is the subject's own act: no one else, including a compromised registry, can produce it. Under managed custody the signature proves only that the registry used the subject's key, not that the subject decided. The evidence of consent is then the registry's audit record of the instruction: the authenticated account and the exact terms approved, or the automatic-approval rule the subject set in advance. A compromised registry could forge both. A registry offering managed custody must say so and must publish each subject's custody mode. The protocol defines both modes; the reference registry currently registers new subjects in managed custody only. The registry operator publishes its own keys, with their history, as a signed key set at `/keys`. AI providers authenticate with a credential issued to them by the registry; end users connect their registry account to a provider once. All parties may be mutually distrustful.

Claim	Who checks	Trusts	Does not prove
Decision: the registry gave this answer to this request	provider, before generating	operator key from the signed directory	that a license exists; that the output is lawful
License: subject and operator agreed to these terms	anyone, offline	subject key as registered at the registry	under managed custody, the subject's own act
Receipt: the provider reports producing this output	registry, rights-holder	provider's key or credential	that the report is true; that every generation was reported
Observation: the provider reports a use without a license	rights-holder	the provider's report	that the use happened as reported; anything granted
Audit log: an event is included under an anchored root	anyone with an inclusion proof	public timestamp calendars	that the log is complete or has no other branch; that what it records is true
Directory: this registry may issue these codes	any client	steward key, pinned origin keys	any right over a person or a work

The principal threat model consists of six scenarios: (1) a licensee fabricating authorization it does not hold; (2) the registry operator fabricating authorization on behalf of a subject; (3) an AI provider generating without consulting the registry; (4) a third party tampering with stored records or the audit log; (5) a bad actor registering as a subject they do not represent; (6) a valid decision replayed for another user, product, or project.

Threat (1) is addressed by two-party signatures. Threat (2) is addressed only in part. Under self custody the subject signature stops the operator from forging a license, but not from signing an allow that names a license that does not exist: the provider verifies the decision, not the license. Without a verifiable license and a trusted binding of the subject's key, such an allow cannot be independently confirmed. An evidence bundle (§10) without a license carrying the subject's signature shows only that the bundle does not support the allow, not that no license ever existed; a

compromised registry may also withhold the bundle, and the subject key's binding is itself the registry's statement. Preventing such an allow, or proving one, needs mechanisms v5 does not specify. Under managed custody a compromised operator can produce both signatures and the record of consent; only the external anchoring of the audit log (§10) makes a later rewrite of that record detectable. Threat (3) cannot be prevented by any protocol; receipts, assertions, and the absence of them provide evidence, not enforcement. Threat (4) is addressed by the audit log: append-only hash chain, per-event signatures, and external anchoring. Threat (5) is addressed by registration trust states and the dispute flow (§4.2). Threat (6) is mitigated by binding the decision to the declared intended use and end user. Accepting one receipt per decision limits what can be reported against a decision, not how many generations a provider runs under it (§13).

4. Entities, Registration, and Cryptographic Material

4.1 Subjects

Every subject is identified by a `subject_id` and a permanent PG code (§5), and is bound to a public signing key registered at issuance. A subject is any entity to which generative rights attach: natural persons (likeness, voice, mannerisms); estates of deceased persons; brands and trademarks; synthetic characters; voice profiles; typefaces; artistic styles; product designs; and creative works whose generative reproduction requires authorization. The protocol does not define what can be a subject — it defines how subjects are registered, how rights are expressed, and how authorization is verified.

4.2 Registration trust

Registering a subject does not by itself prove ownership. A registry records two independent trust axes, and both are visible to verifiers:

Lifecycle — `self_attested`, `pending_verification`, `verified`, `commercial_enabled`, and the restricted states `suspended`, `disputed`, `revoked`, `archived`. Commercial licensing requires both the subject and the account that owns it to be verified.

Authority — `self`, `agency_asserted`, `consented`, `verified`: how well the registrant's right to act for the subject is proven. Every decision carries this value as `subject_authority`, so a provider can see how strong the backing of an allow is.

Which evidence moves a subject between states is registry policy (Layer 3). Subjects flagged as high-risk — minors above all — carry fixed prohibitions that no license can lift (`PG_PROHIBITED_USE`). A registry may add a protection flag on the owner's request; only the registry may remove one.

4.3 Registry operator, providers, and licensees

The registry operator holds an issuer keypair that countersigns licenses and signs decisions, audit events, evidence bundles, and assertions. Key rotation keeps every previous public key in the signed key set, so past signatures stay verifiable.

An AI provider obtains a credential from the registry after proving control of its domain: a sandbox credential, which reaches synthetic test subjects only, and a production credential after a DNS record on that domain. Control of a domain is not proof of who runs the company behind it. The credential is shown once, in the provider's authenticated account; once stored, it is used only on the provider's servers — never in client code, a URL, or a prompt.

A licensee — typically an end user of the provider — does not hold cryptographic material. The user connects their registry account to the provider once (§9); afterwards every query is server-to-server. No protocol material appears in user-facing prompts, so a prompt cannot reveal or replace a credential, and inference APIs stay unchanged. A prompt can still affect how the provider recognizes the subject or what the provider's own agents do; the protocol does not protect against that.

5. Identifiers

Every subject and license has a permanent, human-readable PG code. The format is frozen: a code is cited in licenses, disputes, and court filings, so changing how it is computed would invalidate every code already issued.

Shape	Example	Meaning
PG-<serial><check>	PG-000042*	A subject
PG-<CLASS>-<serial>-<tail><check>	PG-RND-000042-ZZZZZZ1	A license
PG-<ISSUER>-<serial><check>	PG-NWRD-000042=	A subject in another registry

The alphabet is Crockford base32 [5] — digits and uppercase letters without I, L, O, U — so a code survives being read off a screenshot. The serial is zero-padded to at least six digits and widens freely. The license tail is six random characters, so the number of licenses on a subject cannot be inferred. The check character is a position-weighted sum modulo 37 over the body, with five extra symbols (* ~ \$ = !) valid only in the check position; it catches every single-character substitution and every adjacent transposition. A mistyped code is reported as mistyped, distinctly from not found.

Every registry other than the origin one prefixes its codes with exactly four letters from the same alphabet. The prefix is part of the checked body, and bare codes belong to the origin registry forever (§11). Two legacy subject shapes from before the check character (PG-STD-000123, PG-SUB-000123) remain valid. An identifier is resolved by lookup, never parsed for meaning; the issuer prefix is the one thing software may read from it, to know which registry to ask. Anyone can resolve a code without an account through the registry's public resolver.

6. License Schema and Signing

A license body (`pg.license.v2`) contains the subject and licensee identifiers, license class, scope (modalities), default-deny categories, immutable denials, required obligations, contract start and expiry, product and project, purpose, rights-holder rules text, and an extensions array for typed constraints. Optional fields — allowed channels, territories and providers, blocked providers, maximum uses, concurrency limit, granted rights, transferability — are included only when set, so a license signed before a field existed keeps verifying byte for byte. A license is usually bound to one named project; another project is refused (`PG_PROJECT_MISMATCH`).

Classes. RND — research, education, and editorial use; PRM — commercial use; STD — personal use, kept for licenses already issued. Personal use is now a usage record rather than a license (§7), so no new STD licenses are issued. The classes EDT, EST, ENT, PER, and OWN are retired: their identifiers keep resolving, and a retired class is never issued again or given a new meaning.

Extensions express restrictions evaluated at verification time, keyed by versioned type — `pg.temporal_window.v1`, `pg.geo_fence.v1`, `pg.volume_cap.v1`. Unknown extension types fail closed: the engine refuses rather than silently allowing.

Canonical form. Every signed object is serialized as JSON with keys sorted by code point, no whitespace, non-ASCII characters as raw UTF-8, and integers only. Unicode is not normalized: visually identical NFC and NFD strings sign differently, by design. Hashes are SHA-256 in lowercase hex; signatures are Ed25519 over the canonical bytes with the signature field excluded; a key is named by `pg-ed25519:` and the first 32 hex characters of the SHA-256 of the public key.

Issuance is a two-pass operation. The subject signs the canonical body. The registry verifies that signature, allocates the license identifier, and countersigns. Verification reverses the construction; if either signature fails, the license is treated as forged and the verifier returns `PG_INVALID_SIGNATURE`, distinct from `PG_NO_LICENSE` in the audit log.

7. Verification Pipeline

The verification pipeline runs before any output is produced. Once a depiction exists, it can be copied and distributed regardless of any later refusal. The pipeline enforces contextual authorization: not only whether a license exists, but whether the specific intended use falls within its scope.

A provider sends a server-to-server request to `/v1/verify` with its credential, the licensee identifier (omitted for personal use), the subject, a SHA-256 `prompt_hash` — never the prompt text — the model, the modality, and the end user's identifier at the provider. It adds an `intended_use` object: use case (`personal`, `educational`, `research`, `editorial`, `commercial`), channel, product, project, territory, and categories. The engine evaluates the request in a fixed order; each stage is terminal:

Stage 1 — Subject. The subject is resolved; an unknown one returns `PG_NO_SUBJECT`.

Stage 2 — Opt-out. An active opt-out covering the request returns `PG_SUBJECT_OPTED_OUT` immediately, before any caller or license state is consulted. No license can override it.

Stage 3 — Caller. The provider credential and, where a licensee is named, its connection are checked (`PG_NO_PAIR`). A sandbox credential reaches synthetic subjects only (`PG_SANDBOX_SUBJECT_ONLY`).

Stage 4 — Subject status and trust. Pending, paused, disputed, and withdrawn subjects return their own codes. A provider the rights-holder has blocked is refused (`PG_PROVIDER_VETOED`). Fixed prohibitions for high-risk subjects apply (`PG_PROHIBITED_USE`). The trust lifecycle and, for commercial use, the owner's verification are enforced (`PG_SUBJECT_NOT_LICENSEABLE`, `PG_OWNER_NOT_VERIFIED`).

Stage 5 — Personal use. A personal request with no license that passes a fixed floor — images only; no commercial, advertising, endorsement, political, or adult use — is answered `not_blocked` with `PG_STD_TRACKING_ONLY`: nothing prohibits it and nothing is granted. The provider may generate under its own policy and reports the use as an observation (§10); there is no license and no receipt.

Stage 6 — License resolution. Every license for the subject and licensee is checked for status, expiry, contract window, revocation, and both signatures. A license bound to one end user is refused for another (`PG_IDENTITY_MISMATCH`); a license bound to a product or project is refused for a

different one. If a license restricts a dimension that the request omits, the request is refused rather than allowed by omission (PG_MISSING_CONTEXT).

Stage 7 — Scope. Immutable denials are evaluated first (PG_IMMUTABLE_DENIAL); then default-deny categories, the use-case rank of the license class (PG_TIER_VIOLATION), channel, territory, modality, and granted rights (PG_SCOPE_VIOLATION, PG_RIGHTS_NOT_GRANTED).

Stage 8 — Extensions and limits. Typed extensions are evaluated; purchased quantities are reserved atomically (PG_USAGE_LIMIT) and concurrency is capped (PG_CONCURRENCY_LIMIT). A commercial generation of a high-risk subject may be held for a person to decide (PG_HELD_FOR_REVIEW).

Every request returns a signed decision (pg.decision.v1) with a disposition — allow, not_blocked, review, or deny — and a PG_* reason code classified as hard (no retry, reshaped request, or new license changes the answer) or soft. An unknown code is treated as hard. The decision also carries the obligations to apply, the policy_version of the rules that produced it, the subject's revocation epoch, a cache scope (only a plain allow is cacheable, briefly), and the echoed intended use. On PG_NO_LICENSE it adds a remediation link where the user can request a license. All decisions are appended to the audit log.

Checking a decision. A signed allow is not enough on its own. The provider verifies the signature with a key it trusts and then checks that the decision answers its own request. Every echoed member must equal what it sent: subject, provider, licensee, prompt hash, model, modality, and each member of the intended use, including those left empty. The validity window must hold: expires_at is a positive integer, later than issued_at, and not yet passed. If the provider named an end user or an identity link, an allow counts only when it carries provider_user_binding verified and a non-empty identity-link identifier. A correct signature on a decision issued for someone else authorizes nothing. A decision that fails any of these checks is invalid, not a refusal.

8. Opt-Out Registry and Policy Profiles

8.1 Total opt-out

The opt-out registry is structurally distinct from the license registry: it expresses the refusal of permission rather than its grant. An opt-out names a subject, a reason class (personal, estate, safety, legal), a scope (all modalities or a specific one), and an authority proof. The reason text is stored privately and is never returned in any API response or audit log. Registering an opt-out is free.

A registry should make an opt-out harder to register than a license, and publish it as pending during a cooling period: a fraudulent license costs money and leaves a trail, while a fraudulent opt-out would let bad actors lock subjects out of legitimate use. The reference registry uses 14 days. While an opt-out is pending it has no effect on decisions: licenses and personal use continue, and connected providers are notified at once of the date it takes effect. Priority applies from that date. For urgent cases the rights-holder does not wait: pausing the subject takes effect immediately and refuses every request at Stage 4, before any license is consulted. The two differ in who may use them: anyone claiming authority over a subject can file an opt-out, which is why it waits, while only the subject's owner or the registry operator can pause it. The fixed protections for minors and the immutable denials never depend on an opt-out.

Priority has an exact reach. From its effective time every new decision for the subject refuses. A decision issued earlier stays usable until it expires (300 seconds at the reference registry), and a cached plain allow until its cache age runs out, never past its own expiry; a provider learns of the

change from its next decision. Another registry holding the same person is not bound by it in v5 (§11).

8.2 Jurisdiction profiles and legal overrides

The opt-out has priority within one registry's decisions (§8.1); courts and legislatures act outside the protocol. A court may order generation despite an active opt-out. Such an override should be recorded as a parallel, time-limited and scope-limited record, never as a change to the opt-out. Version 5 defines no format, decision, refusal code, or conformance status for it, and the reference registry implements none; a jurisdiction profile would have to specify them.

8.3 Rights-holder controls

Besides the opt-out, a rights-holder may deny specific categories, block specific providers, require review of each request, and attach display-only rules text. These controls are enforced by the pipeline; the rules text itself is shown, not enforced.

9. Provider Credentials and User Connection

Provider onboarding. A provider applies with its identifier, legal name, domain, a security contact, and a contact where rights-holders can reach it. A verified account email on the company's domain yields a sandbox credential; a DNS TXT record on that domain yields a production credential. Credentials are shown once and can be rotated; the old secret stops working immediately.

User connection. Connecting an end user replaces the pair handshake of version 4. The provider sends the user to the registry with a return address registered in advance. The user confirms, and the registry sends back a single-use code valid for five minutes. The provider's server exchanges it, with its credential, for the user's licensee identifier and an identity link. On every later query the provider names that user, and a license the user obtained is bound to that link and answers only for them. An allow for a request that names a user is accepted only with a verified binding to that user (§7). Personal-use queries need no connected user.

A provider becomes available to users other than its own team after it completes one production allow followed by its matching receipt. Connections may be revoked from either side; revocation is delivered by signed webhooks with replay protection and event deduplication.

10. Receipts, Audit Log, Anchoring, and Evidence

Receipts. After an allowed generation the provider files a receipt (`pg.receipt.v2`) naming the decision, prompt hash, output hash, model, and the obligations it applied. The opt-in `pg.receipt.v3` additionally signs the lifecycle event — `preview`, `output_accepted`, `output_delivered`, or `output_published`. A provider with a registered key signs its receipts; the registry stores a hash. Exactly one receipt is accepted per decision, so a replayed decision cannot yield a second one; later stages of the same output — `delivered`, `published` — are recorded against that receipt as lifecycle events, which the registry logs but the provider does not sign. A receipt attests the provider's claim; it does not prove that every generation was reported.

Observations. Uses that need no license are reported as observations (`pg.observation.v1`): personal use answered `not_blocked`, and outputs that the provider's own detectors link to a subject. An observation carries an event identifier, an output identifier, the subject, an output hash, the time, the declared purpose, and one event type: `output.created`, `output.modified` (with the parent output's hash), `output.published`, or `publication.removed` (with a public HTTPS URL,

never fetched by the registry). Events of one output share its output identifier, so an output's history from creation to removal can be followed. A resent event identifier with the same content is accepted once; with different content it is refused. An observation may name a decision only if that decision already has the provider's receipt. The registry signs its audit record of each observation; the observation itself grants nothing and proves only what the provider reported. Observations are the basis for a rights-holder's view of how a subject is used. A `not_blocked` decision has no receipt, so an observation of personal use cannot name the decision that answered it; v5 leaves the two unlinked.

Audit log. Every state-changing operation produces an audit event with an identifier, action, entity, actor, timestamp, metadata, `prev_hash`, event hash, and a signature by the issuer key. The chain gives ordering integrity; the per-event signature gives authenticity even against an attacker with database write access. A Merkle root over event hashes is exposed with inclusion proofs in the manner of Certificate Transparency [4], and roughly once an hour the registry submits the latest signed root to public timestamp calendars outside its control. Submission is best effort: an event is externally anchored only once a later root that includes it has been accepted, a calendar outage leaves a root unpublished, and a certificate states which is the case. Anchoring shows that the log was not rewritten after that moment, never that what it records is true.

Evidence. For any decision the registry returns a signed evidence bundle (`pg.evidence.v1`): the decision, the license it resolved against, every related audit event, an inclusion proof where one exists, and the key set — everything needed to verify it offline, years later.

11. Multiple Registries

PG is an open standard: any company may operate a registry, and compatible registries are listed publicly at pregen.org. To keep their codes from ever colliding, every registry other than the origin one prefixes each code it issues with a four-letter issuer prefix assigned through that public list, and a prefix is never reassigned. A registry that receives a code with a prefix it does not hold answers that the code belongs to another registry and points to the list, instead of reporting it as not found.

The signed directory. A prefix by itself protects nothing: anyone can print one. What gives a namespace meaning is the registry directory at pregen.org, signed by the PRE-GEN steward key. It binds the bare namespace to the origin registry and each prefix to exactly one other registry, together with that registry's API, its key set, and the fingerprints of its operator keys. The API and key-set addresses are HTTPS URLs without a user name, password, query, or fragment. A client accepts the directory only with a valid signature by a steward key it obtained in advance, never from the directory itself. It never accepts a directory older than one it has already seen; a client that keeps the last directory it accepted also refuses a newer one that removes a registry or a key, and a different one with the same sequence number. A license is then valid only when a key listed for the namespace of its code signed it; the rule is the same for every registry, the origin one included. A subject code is resolved only at the registry that owns its namespace. A listing shows which registry may issue codes in a namespace; it does not show that the registry holds any right over a person or a work. Version 5 does not define how fresh a directory must be or how a listed key is revoked.

Pinned keys and announced rotation. The verification libraries pin the origin registry's keys as well, so a stolen steward key cannot add its own key to the bare namespace. A registry announces its next operator key before it signs anything with it: the key is generated and kept offline, and its fingerprint is listed in the directory and pinned in advance, so a planned rotation needs no change at any verifier. Neither measure revokes a stolen key: a stolen operator or steward key keeps producing valid signatures, v5 defines no revoked status for a listed key, and an object's own `issued_at` does not show that it was signed before a theft.

The steward key is held offline by the editor of the standard, not by any registry, and is used only to sign the directory. Its public key is 5cb949aab04186e3e216ec541b847c912fc3f78138c0ec3cb2560b2dad0d1f1b, fingerprint pg-ed25519:dcdd244659e6d0e2426f2b504cb11590. The same values are published in the specification, the repository README, and the DNS TXT record `_pregen-steward.pregen.org`, so that no single channel can substitute them.

Version 5 therefore resolves collisions between codes, not between rights: two registries may hold the same person and answer differently, and v5 defines no rule for which answer prevails. Specified for later versions: directory freshness and revocation of listed keys; one client that reaches every listed registry; opt-outs and protections for minors honoured across registries holding the same person; and a freeze on new licenses while two registries dispute the same subject. The cross-signing, trust delegation, and transparency gossip described in version 4 remain future work.

12. Versioning

A PRE-GEN version is one published release of this specification, numbered as on Zenodo under the concept DOI 10.5281/zenodo.20129901. Versions 1 to 3 were published as Prompt Protocol in May 2026, version 4 as PRE-GEN on 1 June 2026, and this document is version 5. Old material keeps two different kinds of standing. As evidence it stays valid forever: a code keeps resolving to the same thing, and a signed object that verified under the rules and keys of its time still verifies under every later version; retired keys remain published for that reason. As authority it lasts only while current: a decision authorizes only its own request until it expires, a license only within its term and while not revoked or overridden, and a retired key is no longer used to sign; how verifiers refuse new signatures from a compromised key is not yet defined (§11). A version may stop issuing something, never stop recognizing it.

Each signed object carries its own schema marker inside the signed bytes: `pg.decision.v1`, `pg.license.v2`, `pg.receipt.v2` and `v3`, `pg.evidence.v1`, `pg.assertion.v1`. These markers move independently of the PRE-GEN version: adding an optional field never bumps a marker; removing a field or changing its meaning does. Decisions also carry a `policy_version` naming the rules that produced them. Conformance is claimed against named profiles — Verifier, Provider, and Registry — each a list of numbered requirements, with optional features claimed by name. Reproducing every vector is necessary for a claim but not sufficient: the specification lists, requirement by requirement, which ones the vectors and the conformance runner check and which they do not. A run that claims a profile may not skip an operation the profile requires: the runner reports passed, failed, and skipped (not checked) separately, and a required operation that was not checked fails the run. No implementation built by another team from the text alone has yet passed them, so interoperability between independent implementations is specified but not yet demonstrated. Passing results from one team's own implementations, however many, are not that demonstration.

13. Limitations

The protocol does not enforce that AI providers consult the registry. PG is a governance and compliance protocol, not a universal enforcement protocol. Its value is proportional to the size of the compliant ecosystem.

Identity resolution remains an open problem. Name matching catches direct mentions, misspellings, and many obfuscations, but not indirect references, style transfer, or visual resemblance; recognizing a face or a voice is left to the provider's own detectors. PG deliberately leaves face and voice matching to providers: a registry holding biometric templates would become a store of

specially protected biometric data (GDPR Art. 9, Illinois BIPA). A decision covers one subject: an output depicting several subjects cannot yet be authorized as a group. An output hash identifies exact bytes and does not survive re-encoding. A decision is not single-use: one receipt is accepted per decision, but the protocol cannot stop a provider from running several generations under one decision, and lifecycle events after a receipt follow one output, not several outputs or edits made under the same decision. Under managed custody a license proves the registry's use of the subject's key, not the subject's own act (§3).

The protocol does not adjudicate fair use, parody, or other legal exceptions. It refuses by default and provides cryptographic evidence; legal exceptions are left to courts and to jurisdiction profiles, which v5 does not define (§8.2).

14. Implementation Notes

The protocol is defined independently of specific libraries. The published test vectors pin canonical JSON, SHA-256, Ed25519 (RFC 8032 [1]), and the PG code format byte for byte, and a conformance runner replays them against any implementation. User connection uses a one-time authorization code in the manner of OAuth 2.0 (RFC 6749 [2]); webhooks are signed with HMAC-SHA256; the provenance bridge targets C2PA [3]; audit roots are timestamped through OpenTimestamps [6]. The reference registry is PRAMPTA, the first registry operating PG in production. Its TypeScript client is published as `@prampta/sdk` 0.4.0; version 0.7.0, which adds the checks of §7 and a check of the signed directory (§11) that the caller switches on explicitly, is prepared but not yet published. The Python client, available as source, trusts the pinned operator keys of each registry it is configured for and does not read the directory. The steward key is published (§11), the directory at `pregen.org` is signed, and the verification libraries `@pregen/verify` (npm) and `pregen` (PyPI) check decisions and licenses against it with the steward key built in. PRAMPTA is still the only listed registry, so connecting independent registries through the directory is specified but not yet demonstrated. What the reference registry offers today is stated separately in the specification, so that no description of the protocol is read as a claim about one implementation.

15. Ongoing and Future Work

Work on PRE-GEN is ongoing; the specification will be revised as the protocol matures and adopters provide feedback. Planned work includes the multi-registry interfaces of §11 and an explicit chain from decision to generation to output to its lifecycle events for video and multi-step workflows. It also includes group authorization for outputs with several subjects, a training opt-out register for dataset collection, generation guidelines passed from rights-holders to providers, and signed commercial terms and settlement records.

16. Changes Since Version 4

Identifiers. The PG-CLASS-SEQUENCE form is replaced by check-summed subject codes and license identifiers (§5), with four-letter issuer prefixes for other registries.

License classes. STD, RND, PRM; EDT, EST, ENT, PER, OWN retired. Personal use is a usage record answered `not_blocked`; no new STD licenses are issued.

Decisions carry a four-value disposition, policy version, cache scope, revocation epoch, subject authority, and remediation link; the prompt is sent as a hash, never as text.

Callers. Providers authenticate with their own credential and connect end users once; licenses can be bound to an end user, a product, and a project.

Verification is stricter: a decision must match its request exactly, including empty members, carry a valid lifetime, and, when a user is named, a verified binding (§7); a conformance run may not count a skipped required operation as passed (§12); a directory may grow but never lose a registry or key, and registries announce their next key in advance (§11). Signed bytes and all earlier test vectors are unchanged; vectors were added for Unicode key order, integer limits, license countersignatures, and namespaces.

Added: receipts, observations, evidence bundles, assertions, the refusal-code registry with hard and soft codes, registration trust states, published test vectors and a conformance runner, and the public list of registries as a directory signed by the steward key (§11).

References

- [1] S. Josefsson, I. Liusvaara. Edwards-Curve Digital Signature Algorithm (EdDSA). RFC 8032, IETF, 2017.
- [2] D. Hardt. The OAuth 2.0 Authorization Framework. RFC 6749, IETF, 2012.
- [3] Coalition for Content Provenance and Authenticity. Content Credentials Specification v2.1 (under development at ISO as ISO/CD 22144). <https://c2pa.org/specifications/>, 2025.
- [4] B. Laurie, A. Langley, E. Kasper. Certificate Transparency. RFC 6962, IETF, 2013.
- [5] D. Crockford. Base 32. <https://www.crockford.com/base32.html>.
- [6] P. Todd. OpenTimestamps. <https://opentimestamps.org>.
- [7] V. Egorov. PRE-GEN — specification, test vectors, and conformance runner. <https://www.pregen.org>, github.com/Pastheroz/PRE-GEN-site, 2026.
- [8] RSL Technical Steering Committee. Really Simple Licensing (RSL) 1.0. <https://rslstandard.org/rsl>, 2025.
- [9] Human Consent Foundation. Human Consent Standard (HCS) 1.0, draft. <https://humanconsent.org/hcs>, 2026.

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